

MAT 3150 — Homework 3 Hints

These are hints, not finished proofs: each one gives the key idea and leaves the write-up to you.

Problem 1: reverse triangle inequality

Tools.

- Triangle inequality: $|a + b| \leq |a| + |b|$ for all $a, b \in \mathbb{R}$.
- For $c \geq 0$: $|a| \leq c \iff -c \leq a \leq c$.

Hint. Write x as a sum that contains $x - y$:

$$x = (x - y) + y.$$

Apply the triangle inequality to this and rearrange. What upper bound do you get for $|x| - |y|$?

Now do the same thing with the roles of x and y swapped. What do you get for $|y| - |x|$? Note that $|y - x| = |x - y|$.

Finishing. You now have $|x| - |y| \leq |x - y|$ and $-(|x| - |y|) \leq |x - y|$. Which of the tools above turns that pair into the single statement you want?

Problem 2: a limit from the definition

The definition. $a_n \rightarrow L$ means: for every $\varepsilon > 0$ there is $N \in \mathbb{N}$ such that $|a_n - L| < \varepsilon$ for all $n \geq N$.

Step 1: guess L . For large n the “+2” and “+5” are negligible. What is $n/(3n)$?

Step 2: scratch work (don’t hand this in as is). Put $\left| \frac{n+2}{3n+5} - L \right|$ over a common denominator. If your guess is right, the n ’s in the numerator cancel and you are left with a *constant* over something that grows with n .

Then make it simpler by *making the denominator smaller*. For example, $3n + 5 > 3n$, so the fraction gets *bigger* if you replace $3n + 5$ by $3n$. You’re allowed to overestimate here: you only need an upper bound that is still less than ε .

Solve “simple bound $< \varepsilon$ ” for n . That tells you how to choose N (use the Archimedean property to justify that such an N exists).

Step 3: the write-up. Present it in the forward direction:

Let $\varepsilon > 0$. By the Archimedean property choose $N \in \mathbb{N}$ with $N > \frac{1}{\varepsilon}$. Then for all $n \geq N$,

$$\left| \frac{n+2}{3n+5} - L \right| = \dots \leq \dots < \varepsilon.$$

Problem 3: $\infty + \text{bounded} = \infty$

Definitions.

- (y_n) bounded: there is $M > 0$ with $|y_n| \leq M$ for all n , so in particular $y_n \geq -M$.
- $x_n \rightarrow \infty$: for every $K \in \mathbb{R}$ there is N such that $x_n > K$ for all $n \geq N$.
- You must show that for every K there is N such that $x_n + y_n > K$ for all $n \geq N$.

Hint. Using $y_n \geq -M$, find a lower bound for $x_n + y_n$ in terms of x_n and M alone.

Given a target K , how large does x_n need to be so that this lower bound is bigger than K ? The definition of $x_n \rightarrow \infty$ works for *any* threshold, so apply it to that threshold (not to K itself).

Problem 4: averages of an increasing sequence

Let $S_n = x_1 + \dots + x_n$, so $y_n = S_n/n$. You want $y_{n+1} \geq y_n$ for every n .

Approach A: compute the difference. Put $y_{n+1} - y_n = \frac{S_{n+1}}{n+1} - \frac{S_n}{n}$ over the common denominator $n(n+1)$, and use $S_{n+1} = S_n + x_{n+1}$. The numerator should simplify to an expression involving only x_{n+1} , n , and S_n .

Then ask: since (x_n) is increasing, how does each of $x_1, \dots, x_n$ compare to x_{n+1} ? What does that tell you about S_n compared to $n x_{n+1}$?

Approach B: a weighted average. Check that

$$y_{n+1} = \frac{n y_n + x_{n+1}}{n+1},$$

so y_{n+1} is a weighted average of y_n and x_{n+1} . When is a weighted average of two numbers at least as large as the smaller of the two numbers? So what do you need to know about how y_n compares with x_{n+1} ? (An average of numbers is never bigger than the biggest one.)

Side question. Is the hypothesis “positive” actually needed anywhere in your argument? (It’s fine to not use it; just notice.)